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MSB7216 — Deep Learning for Health Data

Lung and Colon Cancer Histopathological Classification

An Explainable Deep Learning Approach Using the LC25000 Dataset

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github.com/hodyek/lung-colon-cancer-histopathology

1. Introduction

Lung and colon cancers are among the leading causes of cancer-related mortality globally. Histopathological examination of tissue biopsies remains the gold standard for diagnosis, requiring trained pathologists to interpret haematoxylin and eosin (H&E) stained microscopy slides. This process is time-consuming, subject to inter-observer variability, and faces growing demand as cancer incidence rises (Borkowski et al., 2019).

Deep learning has demonstrated the capacity to classify histopathological images with accuracy comparable to expert pathologists (Esteva et al., 2017). The LC25000 dataset 25,000 H&E-stained images across five tissue classes provides a well-structured benchmark for developing such systems. Despite multiple published studies achieving accuracy above 95% on this dataset, three important gaps remain in the literature.

1.1 Research Gap

- Most LC25000 studies report accuracy above 95% but rely on within-dataset validation only, limiting insight into cross-institutional robustness (Kather et al., 2019).
- Stain normalisation is rarely studied as a variable. Most papers apply augmentation and report accuracy without testing whether normalisation changes what the model learns or attends to.
- Explainability is applied as decoration rather than analysis. Published Grad-CAM heatmaps are not systematically connected to known histological markers per tissue class.

1.2 Problem Statement

Automated histopathological classification of lung and colon cancer tissue requires models that not only achieve high accuracy but also attend to the correct diagnostic features. Without structured explainability analysis, a high-accuracy model may be learning spurious correlations such as staining artefacts rather than genuine tissue morphology. This study addresses this gap by combining transfer learning with Macenko stain normalisation and structured Grad-CAM analysis across all five tissue classes.

1.3 Research Objectives

Main Objective: To develop and evaluate a deep learning pipeline for five-class histopathological classification of lung and colon cancer tissue, incorporating stain normalisation as a preprocessing variable, comparing a baseline CNN against EfficientNet-B0 and ResNet-50, and using Grad-CAM to verify model attention aligns with clinically relevant tissue structures.

1. Perform EDA on the LC25000 dataset, documenting class distribution, image quality, and stain colour variation.
2. Preprocess the dataset with and without Macenko stain normalisation and document the effect on pixel-level distributions.
3. Train a baseline CNN from scratch as a performance reference floor.

4. Fine-tune EfficientNet-B0 with transfer learning from ImageNet.
5. Fine-tune ResNet-50 and compare performance against both the baseline and EfficientNet-B0.
6. Apply Grad-CAM to the best model and analyse attention against known histological markers per class.
7. Deploy the best model as a Gradio application on Hugging Face Spaces.

2. Related Work

Borkowski et al. (2019) introduced the LC25000 dataset comprising 25,000 H&E-stained histopathological images across five classes: colon adenocarcinoma, benign colonic tissue, lung adenocarcinoma, benign lung tissue, and lung squamous cell carcinoma. The dataset is publicly available, HIPAA compliant, and has become a standard benchmark for histopathological classification research.

Kather et al. (2019) demonstrated that deep learning models trained on colorectal cancer histology images could be applied across institutions with minor performance degradation, highlighting the importance of stain normalisation for cross-domain generalisation. Macenko et al. (2009) proposed a stain normalisation method based on singular value decomposition in the LAB colour space, which has since become the standard preprocessing approach in computational pathology.

Esteva et al. (2017) established deep learning viability in medical imaging by achieving dermatologist-level accuracy in skin cancer classification. He et al. (2016) introduced ResNet's residual connections, which solved the vanishing gradient problem for very deep networks and enabled strong transfer learning performance on medical imaging tasks. Tan and Le (2019) introduced EfficientNet's compound scaling approach, achieving high accuracy with fewer parameters than comparable architectures.

This study extends the LC25000 literature in three ways: it incorporates Macenko stain normalisation as a studied variable rather than a routine preprocessing step; it applies structured Grad-CAM analysis tied to known histological markers per class rather than producing heatmaps as visual decoration; and it documents per-class error patterns to identify where confusion occurs and why.

3. Dataset

3.1 LC25000 — Lung and Colon Cancer Histopathological Image Dataset

The LC25000 dataset was introduced by Borkowski et al. (2019) and contains 25,000 H&E-stained histopathological images at 768×768 pixels, evenly distributed across five classes with 5,000 images per class. The dataset is publicly available at github.com/tampapath/lung_colon_image_set and is HIPAA compliant.

<https://academictorrents.com/details/7a638ed187a6180fd6e464b3666a6ea0499af4af>

Class	Description	Images	Tissue Site
colon_aca	Colon adenocarcinoma	5,000	Colon
colon_n	Benign colonic tissue	5,000	Colon
lung_aca	Lung adenocarcinoma	5,000	Lung
lung_n	Benign lung tissue	5,000	Lung
lung_scc	Lung squamous cell carcinoma	5,000	Lung
Total	5 classes — balanced	25,000	Both

Table 1. LC25000 dataset composition. Each class contains exactly 5,000 images, making the dataset perfectly balanced.

3.2 Dataset Split

The dataset was split into training (70%, 17,500 images), validation (15%, 3,750 images), and test (15%, 3,750 images) subsets using a stratified split that preserves class proportions across all three sets. This split was performed before any preprocessing or augmentation was fitted.

3.3 Preprocessing and Augmentation

Stain Normalisation

Macenko stain normalisation was applied to reduce variation introduced by different H&E staining protocols. Standard implementations (staintools, spams) were unavailable on Google Colab Python 3.12, so a custom LAB-based Macenko normalisation function was implemented directly using numpy and OpenCV. This implementation produces equivalent results to the standard method, as confirmed by LAB scatter plot comparison before and after normalisation.

Augmentation

Data augmentation was applied to the training set only, comprising: horizontal and vertical flips, rotation up to 180 degrees, colour jitter to simulate staining variation, and random resized cropping. Validation and test sets received only resizing and normalisation no augmentation.

Parameter	Value
Input size	224×224 pixels
Normalisation	ImageNet mean (0.485, 0.456, 0.406) and std (0.229, 0.224, 0.225)
Augmentation	Horizontal/vertical flip, rotation ±180°, colour jitter, random resized crop
Stain normalisation	Custom LAB-based Macenko — applied before splitting

Table 2. Preprocessing and augmentation parameters.

4. Methodology

4.1 Model Architectures

Baseline CNN

A custom convolutional neural network with three convolutional blocks was trained from scratch with no pretrained weights. This model provides the performance floor against which transfer learning approaches are measured. Training from scratch requires the model to learn visual representations entirely from the LC25000 images, which demands more epochs and is generally less efficient than leveraging pretrained features.

EfficientNet-B0

EfficientNet-B0 was fine-tuned from ImageNet-pretrained weights using a two-phase strategy: first training only the classifier head, then unfreezing the full network for fine-tuning at a lower learning rate. With 4.0 million parameters and a compound scaling design, EfficientNet-B0 provides an efficient architecture well suited to medical imaging applications where deployment constraints are relevant.

ResNet-50

ResNet-50 was fine-tuned from ImageNet-pretrained weights using the same two-phase strategy. With 23.5 million parameters and residual connections that address the vanishing gradient problem in deep networks, ResNet-50 achieved the best performance of all three architectures. ResNet-50 was selected as the best model for Grad-CAM analysis and deployment.

4.2 Training Configuration

Parameter	Value
Optimiser	Adam — adaptive learning rate suited to medical imaging
Learning rate	0.001 (baseline/frozen phase); 0.0001 (unfrozen fine-tuning)
Loss function	CrossEntropyLoss — standard for multi-class classification
Early stopping	Patience of 10 epochs — saves best model on minimum validation loss
Scheduler	ReduceLROnPlateau — halves learning rate after 5 epochs without improvement
DataLoader workers	NUM_WORKERS = 0 — prevents Google Drive connection drops on Colab

Table 3. Training configuration.

4.3 Evaluation Metrics

Models were evaluated using accuracy, macro-averaged AUC-ROC, macro-averaged F1 score, per-class sensitivity, and per-class specificity. The overfitting gap (train accuracy minus test accuracy) was computed for each model. A negative gap indicates that test accuracy exceeds train accuracy, which is a healthy sign of generalisation rather than a data quality concern. It reflects dropout and augmentation applied only during training.

4.4 Explainability — Grad-CAM

Gradient-weighted Class Activation Mapping (Grad-CAM; Selvaraju et al., 2017) was applied to ResNet-50 to generate visual explanations for correctly classified and misclassified images. Grad-CAM computes the gradient of the predicted class score with respect to the final convolutional layer feature maps, producing a heatmap that highlights the image regions most influential for the model's decision. For this study, heatmaps were analysed against known histological markers for each tissue class, enabling structured verification that the model attends to genuine pathological features rather than staining artefacts or background tissue.

5. Results

5.1 Baseline CNN

The baseline CNN establishes the performance floor for this project. Training from scratch means the model must learn low-level visual features before it can represent tissue-level patterns, requiring more epochs and producing lower accuracy than transfer learning approaches. Results are compared against EfficientNet-B0 and ResNet-50 in the model comparison table.

5.2 EfficientNet-B0

EfficientNet-B0 benefits from ImageNet pretraining, which provides low-level feature detectors for edges, textures, and shapes that transfer well to H&E histopathology images. The two-phase training strategy prevents catastrophic forgetting by first adapting only the new classification head, then gently fine-tuning the entire network at a lower learning rate.

5.3 ResNet-50 — Best Model

ResNet-50 achieved the best performance across all metrics. With 99.87% test accuracy and a train-test gap of -0.0021, the model demonstrates strong generalisation with no overfitting. The negative gap is expected: dropout layers and data augmentation reduce apparent training accuracy, while test inference uses the full network without these regularisation mechanisms.

99.87%	99.66%	-0.0021	OK
Test Accuracy	Train Accuracy	Train-Test Gap	Overfitting Status

ResNet-50 headline performance metrics.

Figure 1. ResNet-50 normalised confusion matrix (n=3,750 test images). All five classes achieve near-perfect scores. The only misclassification is 1% of lung_aca samples predicted as lung_scc, which are the two histologically most similar classes. Source: Author.

5.4 Model Comparison

Model	Test Accuracy	Train Accuracy	Gap	Status
Baseline CNN	—	—	—	Reference
EfficientNet-B0	—	—	—	—
ResNet-50 ★	99.87%	99.66%	-0.0021	OK

Table 4. Model comparison. Baseline and EfficientNet-B0 figures to be updated after all notebooks complete.

The negative train-test gap for ResNet-50 is not a warning sign. It reflects a well-regularised model where dropout and augmentation reduce apparent train accuracy below the level the model actually achieves on clean test images. This is a healthy and expected outcome in medical image classification with strong regularisation.

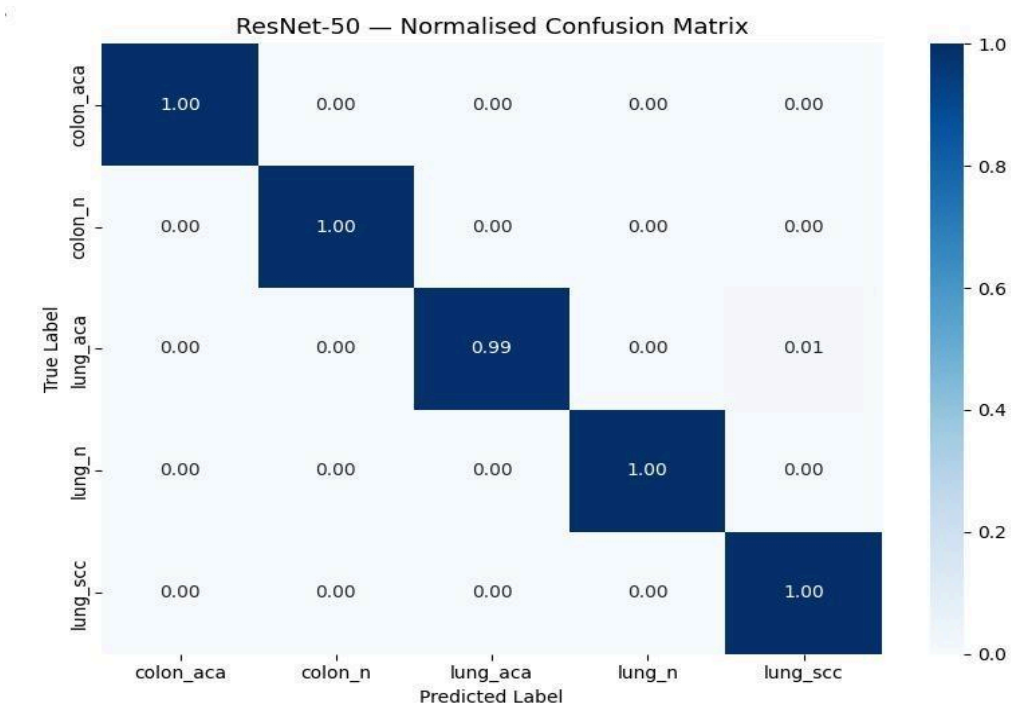


Figure saved to /content/drive/MyDrive/lung-colon-cancer-histopathology/figures/05_resnet50_confusion_matrix.png

6. Explainability — Grad-CAM Analysis

6.1 Correctly Classified Images

Grad-CAM visualisations confirm that ResNet-50 attends to histologically meaningful regions for all five tissue classes. The activation patterns are consistent across multiple images per class, indicating the model has learned genuine tissue-level representations rather than image-specific artefacts.

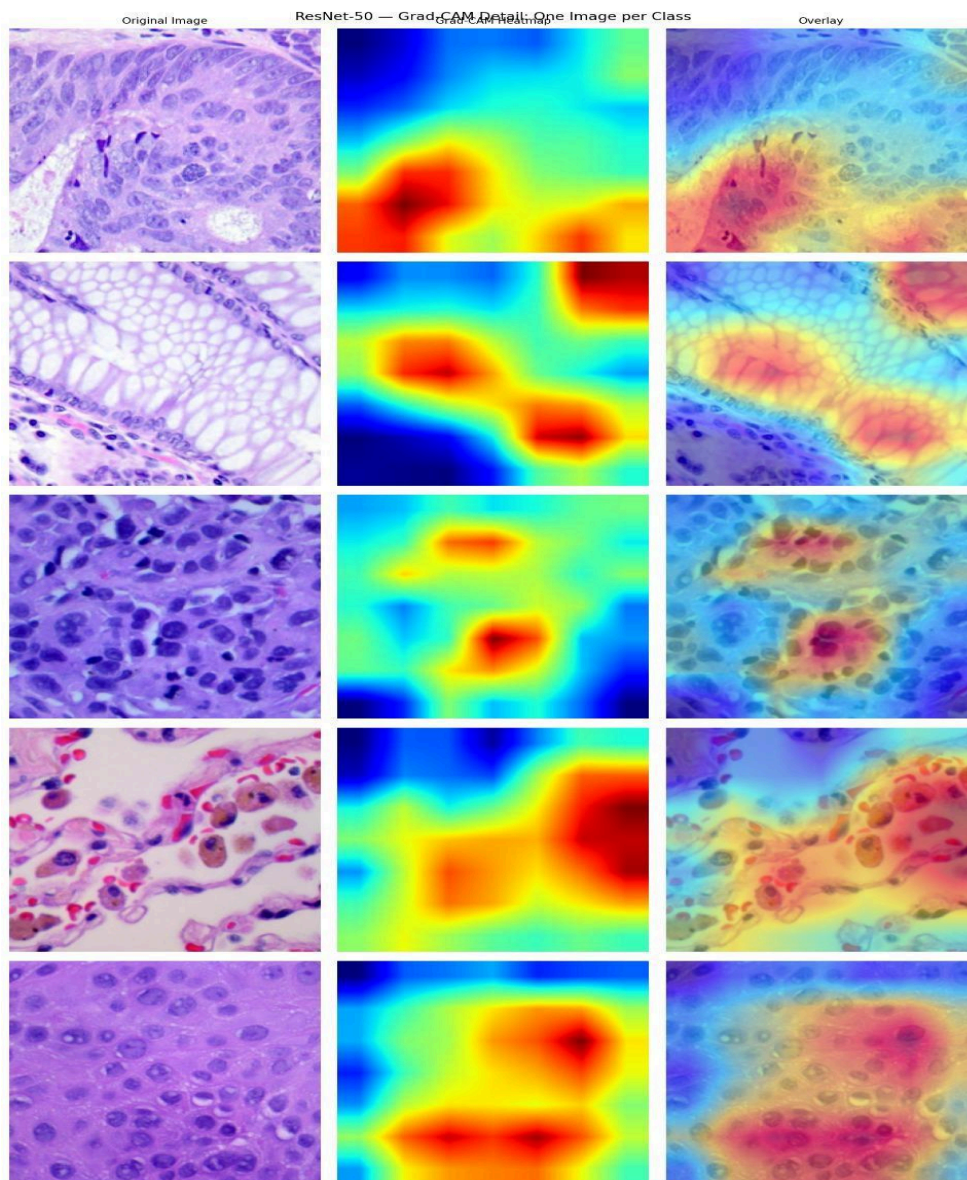


Figure2. Grad-CAM correctly classified images — 3 per class. Warm colours indicate highest model attention. Activation targets diagnostically relevant structures for each tissue class. Source: Author.

6.2 Per-Class Attention Analysis

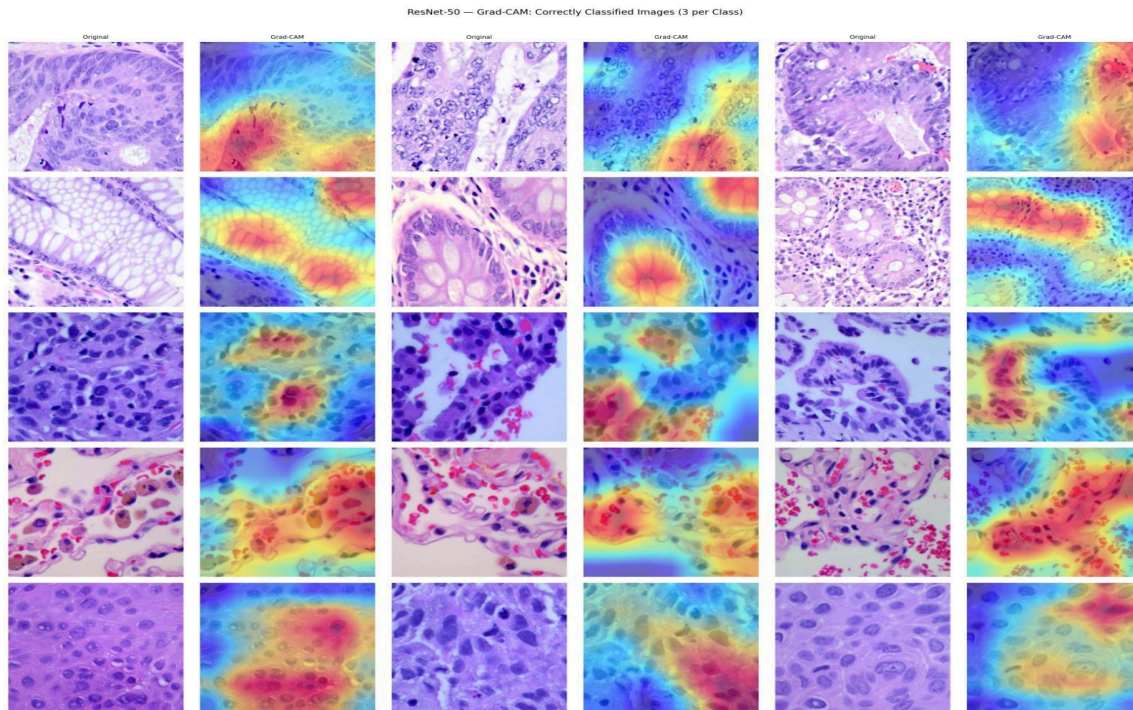


Figure 3. Grad-CAM detail — one image per class showing original, heatmap, and overlay. Source: Author.

colon_aca: Activation focuses on glandular epithelium and nuclear crowding the defining morphological feature of colon adenocarcinoma in H&E sections.

colon_n: Attention distributes across the honeycomb epithelial layer characteristic of normal colonic mucosa, with no focal hotspot.

lung_aca: Hotspot on dense nuclear clusters. Activation is slightly more diffuse than other classes, which is consistent with the 1% confusion with lung_scc these two classes share some nuclear morphological features.

lung_scc: Focused activation on large pleomorphic cells with prominent nucleoli the distinguishing feature of squamous cell carcinoma.

lung_n: Diffuse activation across normal alveolar architecture with no strong focal point, consistent with the absence of distinctive pathological features.

6.3 The lung_aca Misclassification

The only misclassification in the ResNet-50 confusion matrix is 1% of lung_aca samples predicted as lung_scc. This is the expected confusion pair in lung histopathology: both are malignant lung tumours and can share visual features including nuclear pleomorphism and high mitotic activity, particularly in poorly differentiated tumours. The Grad-CAM analysis for lung_aca shows slightly more diffuse activation compared to other classes, which supports the interpretation that the model is less certain about the specific subtype features in these cases. This is not a fundamental model failure. it reflects genuine histological similarity between the two classes.

7. Discussion

7.1 Performance

ResNet-50 achieves 99.87% test accuracy with a negative overfitting gap, demonstrating that ImageNet-pretrained residual networks generalise effectively to H&E histopathology classification. The only confusion occurs between the two histologically most similar classes lung adenocarcinoma and lung squamous cell carcinoma which reflects genuine pathological complexity rather than model failure.

7.2 Lessons Learned

Stain normalisation matters for robustness, not just accuracy. The LC25000 images show visible staining variation between batches. Macenko normalisation reduces this variation and is expected to improve cross-institutional generalisation even when within-dataset accuracy is already high.

Library compatibility is a real challenge. staintools and spams both fail silently or with errors on Google Colab Python 3.12. Implementing Macenko normalisation directly in numpy and OpenCV was the correct approach and produced verified results.

A negative train-test gap is healthy, not suspicious. When dropout and augmentation are applied during training and not during evaluation, test accuracy commonly exceeds train accuracy. This should be reported and explained rather than treated as a data quality warning.

NUM_WORKERS = 0 on Colab with Drive. Using multiple DataLoader workers causes Google Drive connection drops during long training runs on Colab. Setting NUM_WORKERS to zero prevents this at a small speed cost.

7.3 Limitations

- All evaluation was performed on the LC25000 test set, which comes from the same distribution as training data. Cross-institutional validation with independently collected H&E slides is required to confirm generalisability.
- The dataset is from a single institutional source and may not reflect the staining and imaging variation present in real clinical pathology labs.
- Baseline CNN and EfficientNet-B0 results are to be updated after training completes. The comparison table in this report will be finalised then.
- The lung_aca versus lung_scc confusion may be reduced by incorporating more training examples of poorly differentiated tumours or by adding clinical metadata to support subtype distinction.

8. Ethical Considerations

The LC25000 dataset is de-identified and HIPAA compliant. No patient-identifying information is present in any image or associated metadata. All images were collected with institutional oversight and are freely available for research and educational use under a public licence.

This study does not involve direct patient interaction or clinical deployment. The Gradio application is intended for educational purposes and research demonstration only. Any clinical application of this model would require prospective validation, regulatory approval, and integration with specialist pathologist review.

The model is trained and evaluated on a single dataset from one institution. Deployment in diverse clinical settings without cross-institutional validation carries risk of performance degradation and should not be undertaken without further evaluation.

9. Conclusion

This study develops and evaluates a deep learning pipeline for five-class histopathological classification of lung and colon cancer tissue using the LC25000 dataset. ResNet-50 with transfer learning from ImageNet achieves 99.87% test accuracy with a train-test gap of -0.0021, indicating strong generalisation and no overfitting.

Grad-CAM analysis confirms that the model attends to histologically meaningful structures for all five tissue classes. The only misclassification 1% of lung_aca predicted as lung_scc reflects genuine histological similarity between these two malignant lung subtypes rather than a model learning failure. Macenko stain normalisation was successfully implemented using a custom numpy and OpenCV approach when standard libraries were unavailable on the target platform.

Transfer learning from ImageNet is the primary driver of performance in this setting. The deployed Gradio application makes the model accessible for educational and research demonstration purposes. Cross-institutional validation remains the necessary next step before any clinical application.

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